

Day 21 - Lecture 21.9: Scalar Multiplication & Linear Scaling

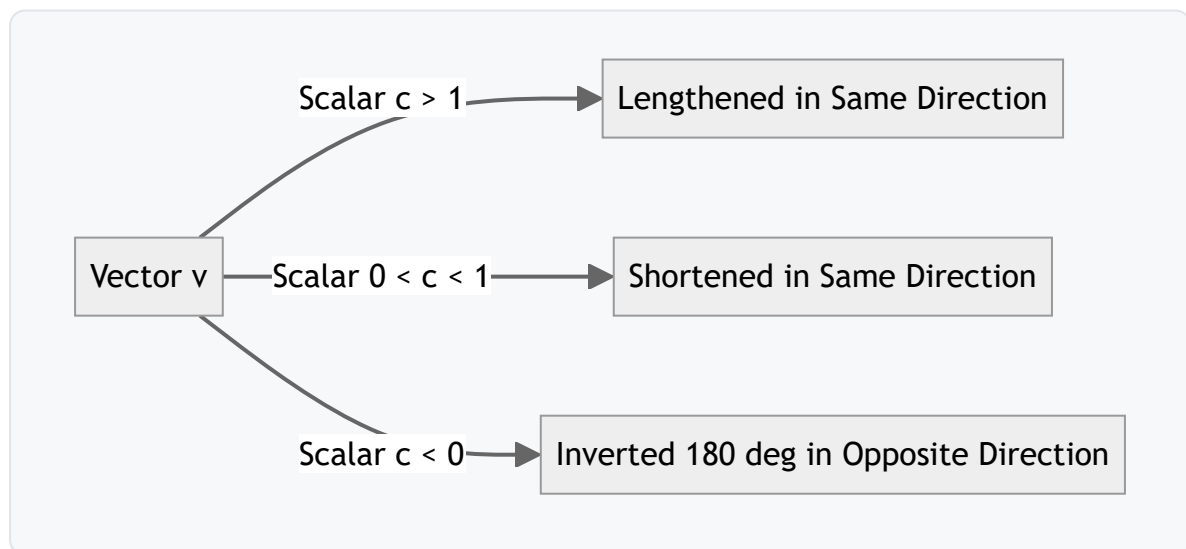
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1. Overview & Operational Scaling

Scalar Multiplication scales a vector by a real scalar number $c \in \mathbb{R}$. In Machine Learning optimization, scalar multiplication directly governs the step size of **Gradient Descent**:

$$\boxed{\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_t)}$$

where the learning rate η is a scalar multiplier that scales the gradient vector $\nabla \mathcal{L}$.



2. Mathematical Formalism

Component-wise Definition:

For scalar $c \in \mathbb{R}$ and vector $\mathbf{v} \in \mathbb{R}^n$:

$$c \cdot \mathbf{v} = c \begin{bmatrix} v_1 & v_2 & \vdots & v_n \end{bmatrix} = \begin{bmatrix} c \cdot v_1 & c \cdot v_2 & \vdots & c \cdot v_n \end{bmatrix}$$

Norm Scaling Law:

The magnitude of the scaled vector scales by the absolute value of the scalar:

$$|c \cdot \mathbf{v}|_2 = |c| \cdot |\mathbf{v}|_2$$

Directional Properties:

- If $c > 0$: Preserves orientation (angle = 0°).
 - If $c = 0$: Collapses the vector to the null vector $\mathbf{0}$.
 - If $c < 0$: Inverts orientation completely (angle = 180°).
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3. Python Implementation

```
import numpy as np

v = np.array([3.0, 4.0])
c_pos = 2.5
c_neg = -1.5

v_scaled_pos = c_pos * v
v_scaled_neg = c_neg * v

print(f"Original v: {v}, ||v|| = {np.linalg.norm(v):.2f}")
print(f"Scaled by {c_pos:4.1f}: {v_scaled_pos}, ||c*v|| = {np.linalg.norm(v_scaled_pos):.2f}")
print(f"Scaled by {c_neg:4.1f}: {v_scaled_neg}, ||c*v|| = {np.linalg.norm(v_scaled_neg):.2f}")
```

4. Key Takeaways & Interview Points

- **Learning Rate as Scalar:** If η is too large, scalar multiplication causes gradient steps to explode; if η is too small, steps become imperceptible.
- **Span of a Vector:** The set of all scalar multiples $c\mathbf{v} \mid c \in \mathbb{R}$ forms a 1D continuous subspace (a line through the origin).